Sahaj Bhadja

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Summary

Software Engineer with 2+ years building **production AI applications**, **retrieval systems**, secure APIs, and full-stack workflows using Python, TypeScript, React, and cloud infrastructure.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Advanced NLP, Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Data Structures, Software Engineering, Object-Oriented Programming, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Built fault-tolerant FastAPI inference workflows with asynchronous workers, explicit job states, failure isolation, and monitoring, processing 1K+ AI jobs at 3x throughput across **reliable distributed production services**.
- Architected a multi-tenant Azure SQL and cloud-storage platform for 500+ users, reducing dashboard latency 71% through **secure data isolation, indexed retrieval, validation, and performance-focused backend optimization in production**.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Developed React and Flask experiences for uploads, model selection, grading history, and feedback, supporting 10K+ daily requests through **production APIs, cross-functional iteration, and end-to-end feature ownership with users**.
- Engineered concurrent Python ingestion with ThreadPoolExecutor, OCR, timeouts, and multimodal model fallbacks, processing 1K+ files per batch while reducing turnaround 60% through **testing, evaluation, and fault isolation**.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built a PyTorch Decision Transformer for offline reinforcement learning, processing 100K+ trajectory logs through reproducible pipelines and improving planning 18% through **model training, evaluation, and performance experimentation**.
- Integrated HD-map geometry, lane direction, and ego trajectories through spatial-temporal encoding and multi-head attention, improving closed-loop stability 15% while **debugging complex model and simulation failures during controlled experiments**.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Optimized distributed APIs and data pipelines, improving response times 30% through object-oriented patterns, AWS Glue, S3, caching, and **systematic profiling across cloud and cross-platform execution environments at scale**.
- Automated builds and 20+ unit and integration suites across Windows, Linux, and macOS using Jenkins, PyTest, Mockito, and virtual machines, strengthening **CI/CD, regression coverage, and release reliability**.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Built eight modular services for authentication, resources, billing, permissions, and notifications, synchronizing 2M+ records through **REST APIs, WebSockets, React interfaces, and secure real-time distributed workflows across production environments**.
- Automated Docker and Jenkins delivery, reducing deployment time 50% while applying tests, Sentry, Azure Monitor, and Kibana for **observability, troubleshooting, and maintainable production operations reliably across services**.

Skills

Languages & Frontend: Python, TypeScript, JavaScript, Java, Go, SQL, React, Next.js, HTML5, CSS3
AI & Retrieval: PyTorch, Transformers, LLMs, Agentic AI, RAG, Embeddings, FAISS, Pinecone, Hybrid Search, Evaluation
Backend & Data: FastAPI, Flask, REST APIs, Microservices, PostgreSQL, Azure SQL, Redis, Vector Databases, Async Processing
Cloud & Security: AWS, Azure, GCP, Kubernetes, Docker, CI/CD, Jenkins, GitHub Actions, Git, Authentication, RBAC, Audit Logging
Engineering: Distributed Systems, API Design, Unit Testing, Integration Testing, Code Reviews, Debugging, Observability, Performance Profiling, AI Coding Tools

Projects

MatterGraph | *Python, TypeScript, PostgreSQL, Vector Search, Knowledge Graphs*

- Designed a **secure document-intelligence platform** combining hybrid retrieval, entity relationships, source citations, role-based access, redaction, and evaluation datasets to support reliable analysis across large professional-services matter collections.

AgentWorkbench | *Python, TypeScript, React, LLMs, FastAPI, PostgreSQL*

- Built an **agentic workflow system** with typed tools, approval gates, context management, parallel execution, audit logs, and regression evaluations for secure, observable automation of complex knowledge-work tasks.

Big-Brain | *LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP, REST APIs* | Live

- Built a **semantic retrieval and RAG platform** over 50K+ documents using LLaMA-2, FAISS, and Pinecone, enabling contextual question answering and summarization with sub-second retrieval through optimized embedding pipelines.